

# Labour Supply 2: Labour Supply and Public Policy

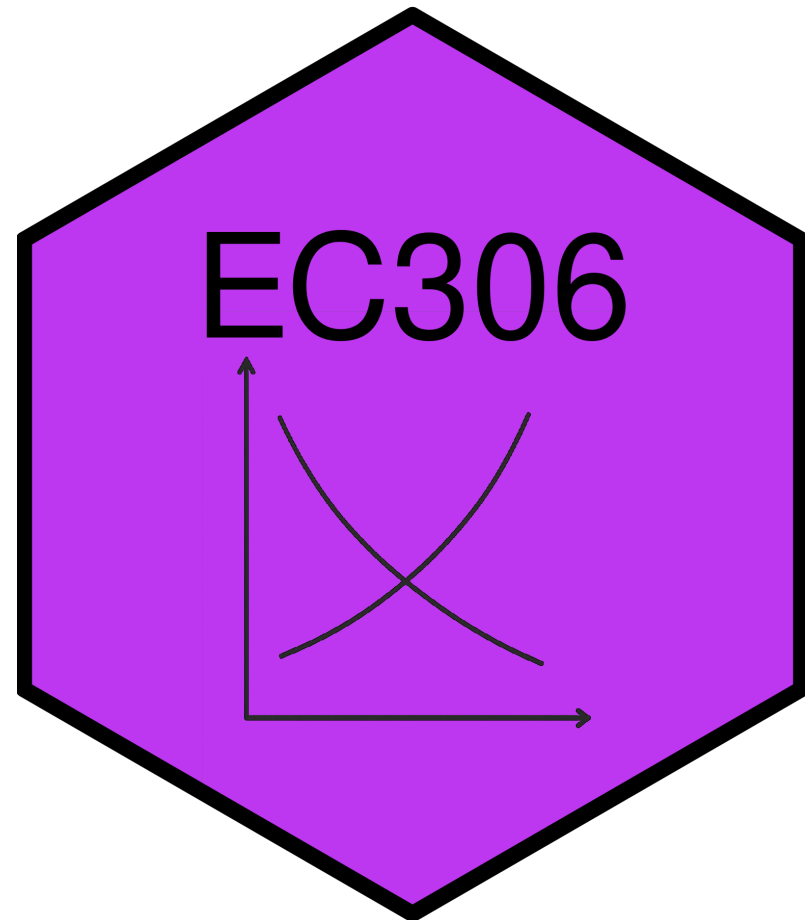
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EC306- Wages and Employment

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# Goals of This Section

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In this section we will:

- Discuss various public policies that affect labour supply
  - Taxes/subsidies, transfers, employment insurance, worker compensation
- Show effect of these policies on work decisions using economic models
- Examine statistical evidence for the model predictions

# Income Maintenance Programs

# Overview of Income Maintenance Programs

- Represent about 10% of GDP in Canada
- Primary aims:
  - Raise income of specific groups (parents, elderly)
  - Supplement low wages
- Major programs: child benefits, employment insurance, social assistance, pensions

## ! Key Challenge

Balancing income support with work incentives

# Government Transfers in Canada

In 2017, 83% of families received some form of government transfer:

| Program                | % Receiving | Average Amount |
|------------------------|-------------|----------------|
| Federal Child Benefits | 20.2%       | \$6,314        |
| Employment Insurance   | 14.0%       | \$8,352        |
| Social Assistance      | 11.3%       | \$8,367        |
| Old Age Security       | 25.1%       | \$8,566        |
| CPP/QPP                | 32.4%       | \$10,092       |

# Issues in Program Design

## Universal vs. Targeted Programs

- Universal: easy to implement but expensive
- Targeted: benefits only those in need but costly to administer

## Work Disincentives

- Some programs reduce incentive to work
- Example: welfare benefits can encourage labour force dropout for those with strong preference for leisure

## Permanent vs. Transitory Income Loss

- Different situations require different program designs
- Temporary layoff → short-term support
- Long-term disability → longer support + reintegration costs

# Demogrants, Welfare, Wage Subsidies

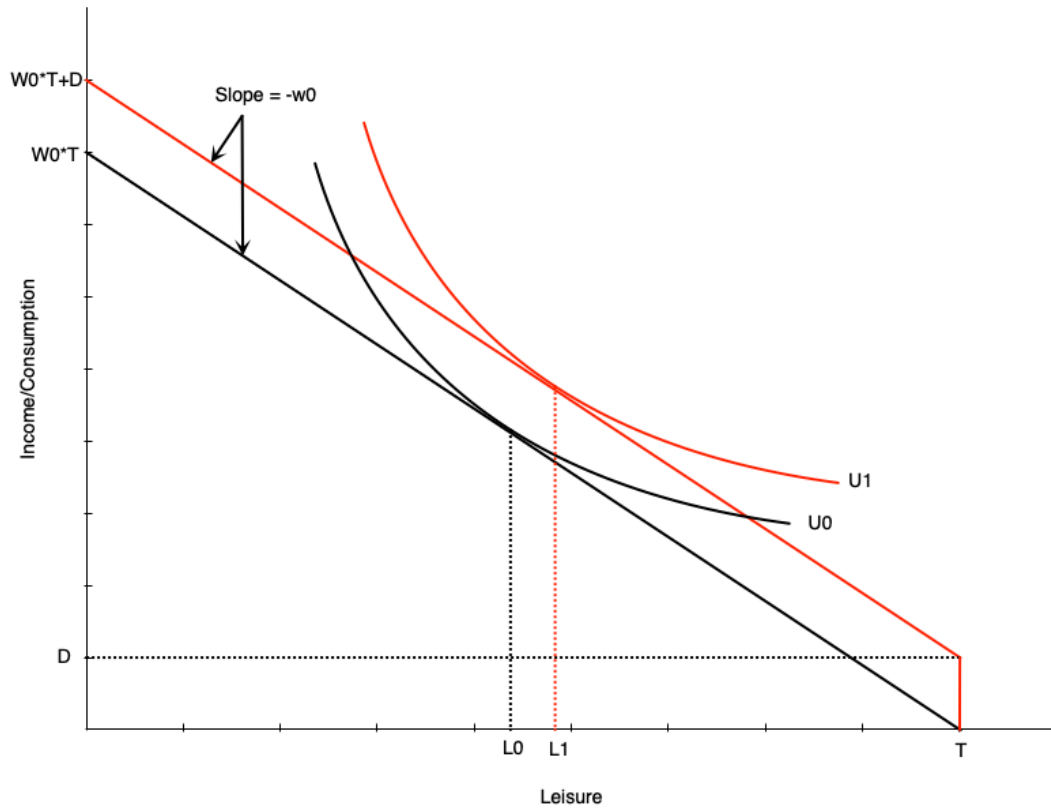


# Four Standard Income Support Programs

We will examine four policy approaches:

1. **Demogrants** - Fixed payment to all regardless of income/employment
2. **Welfare** - Lump-sum payment for non-workers, reduced dollar-for-dollar with earnings
3. **Negative Income Tax (NIT)** - Welfare payment that decreases at rate  $< 100\%$  with earnings
4. **Wage Subsidies** - Addition to the wage rate

# Demogrant: Theory



How it works:

- Unconditional lump sum transfer  $D$
- Shifts budget line up parallel
- Wage unchanged → pure income effect

Labor supply impact:

- If leisure is normal → optimal leisure rises
- Reduces hours worked
- May induce non-participation

## ⚠ Work Incentive Effect

Unambiguously negative

# Demogrant: Canadian Example

## Universal Child Care Benefit (UCCB)

- Paid to families with children under 18
  - \$160/month per child under 6
  - \$60/month per child aged 6-17
- Replaced in 2016 by Canada Child Benefit (CCB)
  - More targeted to low/middle income families

### Evidence (Schirle, 2015)

UCCB reduced labour supply of women with children under 6 by ~3.2 percentage points for low-educated married women

# Welfare: Overview

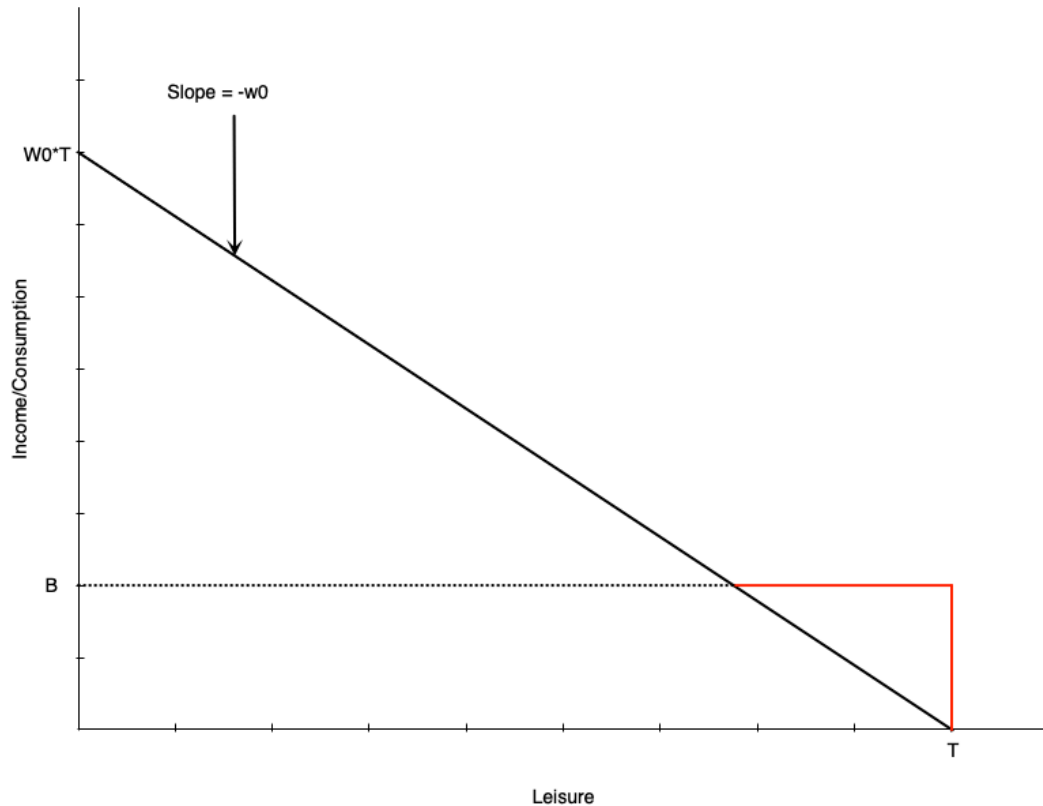
Welfare = Social/Income Assistance in Canada

- Lump-sum payment to non-workers
- Reduced **dollar-for-dollar** with earned income

## ! Historical Criticism

Creates negative work incentives for some, but must balance against need for income support

# Welfare: Budget Constraint



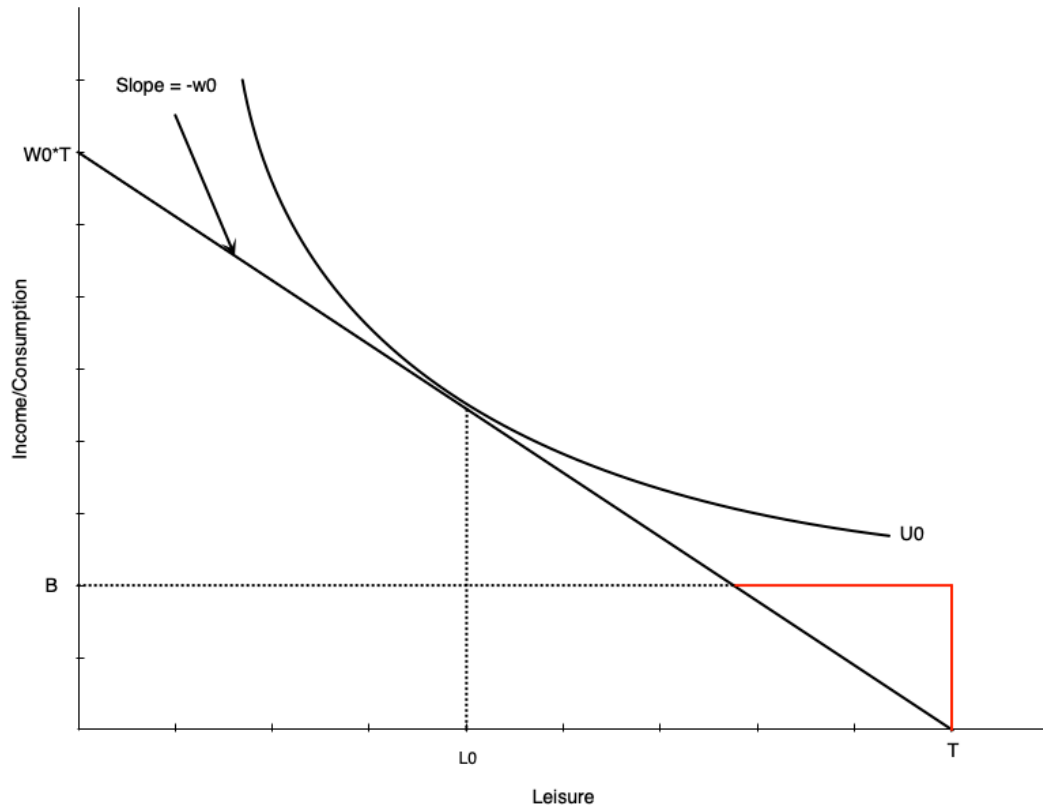
## Budget line structure:

- Black line: original budget (leisure at  $L_0$ )
- Red then black: with welfare benefit  $B$

## Key features:

- Benefit  $B$  at zero hours of work
- Reduced \$1 for each \$1 earned
  - Equivalent to 100% tax rate on earnings
- Income constant at  $B$  until earned income equals benefit
- After that, earn wage  $w_0$  per hour worked

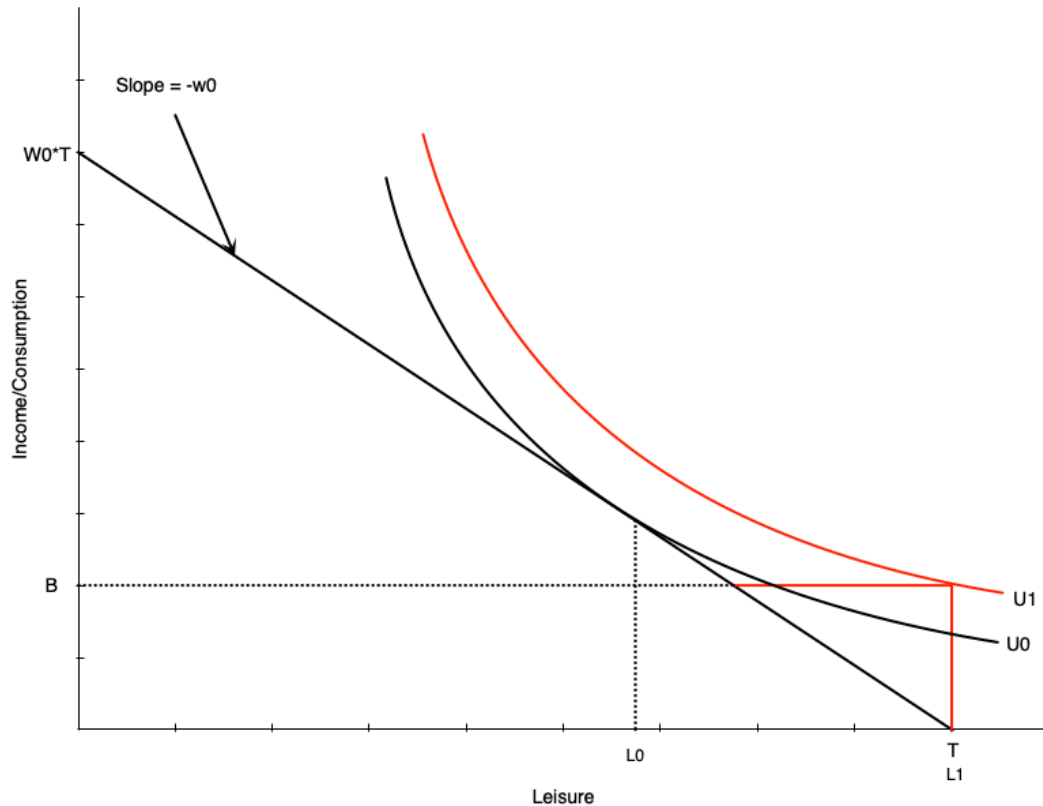
# Welfare: No Effect Case



For some workers, welfare has no effect:

- Person working  $H_0$  hours before benefit
- Still maximizes utility at  $H_0$  hours after
- Generally occurs with people already working many hours

# Welfare: Work Disincentive Case



Others face strong work disincentive:

- Person chooses **zero hours** after welfare benefit

This occurs when:

- Welfare benefit is generous
- Individual places high value on leisure

## Warning

This is the problematic effect of welfare programs

# Combating Work Disincentives

## Policy options:

### 1. Reduce benefit level

- But may not provide adequate support

### 2. Increase wages

- Through minimum wages, subsidies, training, unions
- Can be expensive and may reduce labour demand

### 3. Change preferences

- Alter attitudes toward work/leisure
- Difficult to achieve

### 4. Negative income tax

- Reduce tax-back rate on earnings
- Discussed next...



# Welfare: Evidence from Quebec

**Natural experiment:** Welfare benefit structure changed in 1989

**Before 1989:**

- Age-based benefits for singles/couples with no kids
- Under 30: \$185/month
- 30 and over: \$507/month

**Results at age 30:**

- Much longer time on welfare
- Lower employment rates

## Note

Turning 30 created a **large discontinuous jump** in benefits

# Negative Income Tax: Overview

## What is NIT?

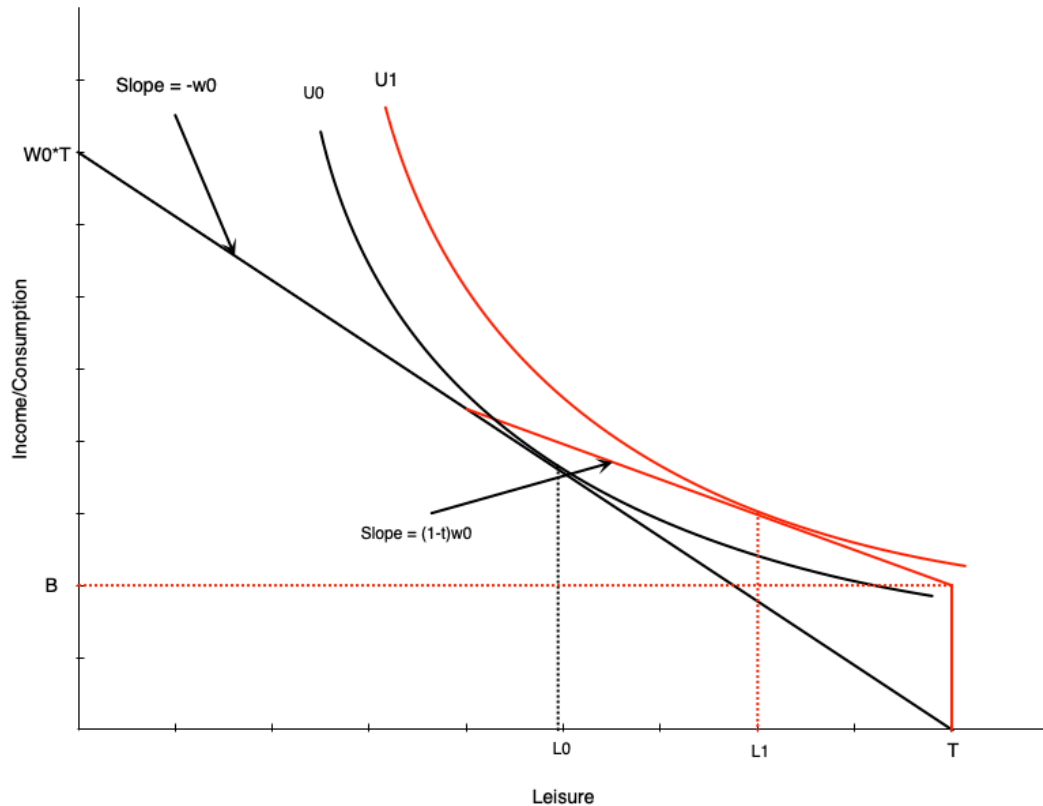
- Welfare program where benefit reduced at rate  $< 100\%$  of earnings
- Example: \$10,000 benefit reduced by 50¢ per dollar earned

### Advantages

Provides income support with fewer work disincentives. Still maintains incentive to work since benefit reduces slowly.

Also known as: Guaranteed Annual Income (GAI) or Basic Income

# Negative Income Tax: Budget Constraint



NIT has two components:

- Benefit  $B$  at zero hours of work
- Tax rate  $t$  on earned income (until earned income equals benefit)

Effect on labour supply:

- Pivots budget line up vs. pure welfare
- Work hours reduced from  $H_0$  to  $H_1$
- Person still works some hours

❗ Important

Work disincentive less than pure welfare

# Negative Income Tax: Income and Substitution Effects

## Income effects:

- Benefit increases non-labour income  $\rightarrow$  increases leisure  $\uparrow$
- Tax has negative income effect  $\rightarrow$  reduces leisure  $\downarrow$
- **Net:** benefit effect dominates  $\rightarrow$  leisure  $\uparrow$

## Substitution effect:

- Tax rate reduces effective wage
- Makes leisure relatively cheaper  $\rightarrow$  reduces work hours  $\downarrow$

### Overall Effect

Negative effect on work hours, but **smaller than pure welfare**

# Negative Income Tax: Math

The budget constraint with NIT:

$$C = \begin{cases} wh + B - twh, & \text{if } B - twh \geq 0 \\ wh, & \text{otherwise} \end{cases}$$

Interpretation:

- Consumption = earned income ( $wh$ ) + net benefit ( $B - twh$ ) if net benefit  $> 0$
- Otherwise just earned income ( $wh$ )

**Break-even point:** benefit reaches zero when  $wh = \frac{B}{t}$

# MINCOME: Manitoba's NIT Experiment

## Program details (1970s):

- Randomized controlled trial
  - Treatment group received NIT
  - Control group did not
- Conducted in Dauphin (saturated - everyone got NIT) and Winnipeg
- 7 treatment groups with different benefit levels and tax rates
- Ended in 1979 due to political changes

# MINCOME Treatment Groups

|                        |        | Tax Rate (on total Income) |        |        |
|------------------------|--------|----------------------------|--------|--------|
|                        |        | 35%                        | 50%    | 75%    |
| Guarantee at enrolment | \$3800 | Plan 1                     | Plan 3 | *      |
|                        | \$4800 | Plan 2                     | Plan 4 | Plan 7 |
|                        | \$5480 | X                          | Plan 5 | Plan 8 |
|                        |        | Plan 9                     |        |        |

\* Plan 6 was collapsed into Plan 7 because participants viewed this option as too punitive and left the experiment.  
X This option was deemed to expensive for the budget.  
Plan 9 was the control group

Source: Wikipedia



# MINCOME: Results

## Labour supply effects:

- Small decline in hours worked
  - Mainly mothers of newborns
  - Women in two-parent families
  - No significant decline for men

## Health outcomes:

- Fewer hospitalizations and doctor visits
- Especially for mental health and accidents

## Education:

- Increase in adolescents finishing high school



# Wage Subsidies: Theory

## What is a wage subsidy?

- Payment to workers that increases their wage rate
- Example: \$5/hour subsidy increases \$15/hour wage to \$20/hour

## Effect on budget constraint:

- Budget line pivots upward

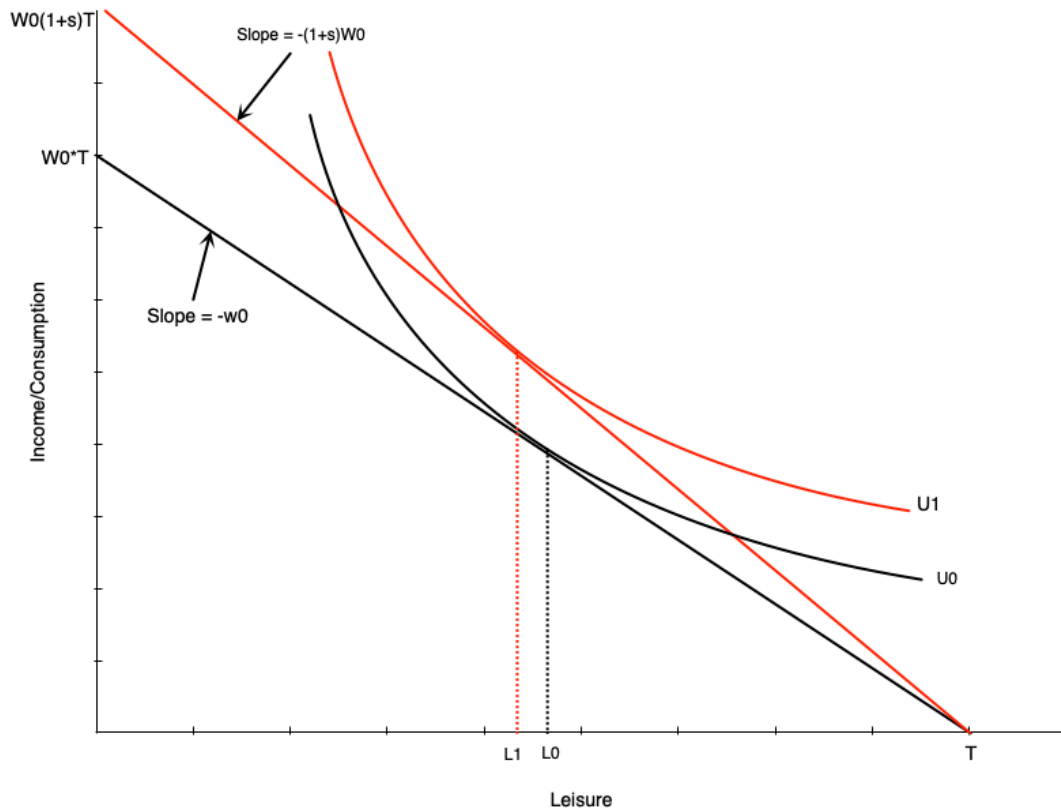
## Labour supply effects:

- Income effect → increases leisure ↑
- Substitution effect → decreases leisure ↓

### Warning

Overall effect on work hours is **ambiguous**

# Wage Subsidies: Diagram



Budget line pivots up with subsidy

- Graph shows person working more
- But effect is theoretically ambiguous
- Income and substitution effects work in opposite directions

# Wage Subsidies: Policy Implications

## Advantages

- Less negative work incentives than welfare or NIT
- No reduction in effective wage rate

## Disadvantages

- Do not provide income support to non-workers
- May not help those in most need

## In Canada:

- Pure wage subsidies mostly go to **employers**
- Unclear if any benefits passed on to workers

# Tax Credits: Overview

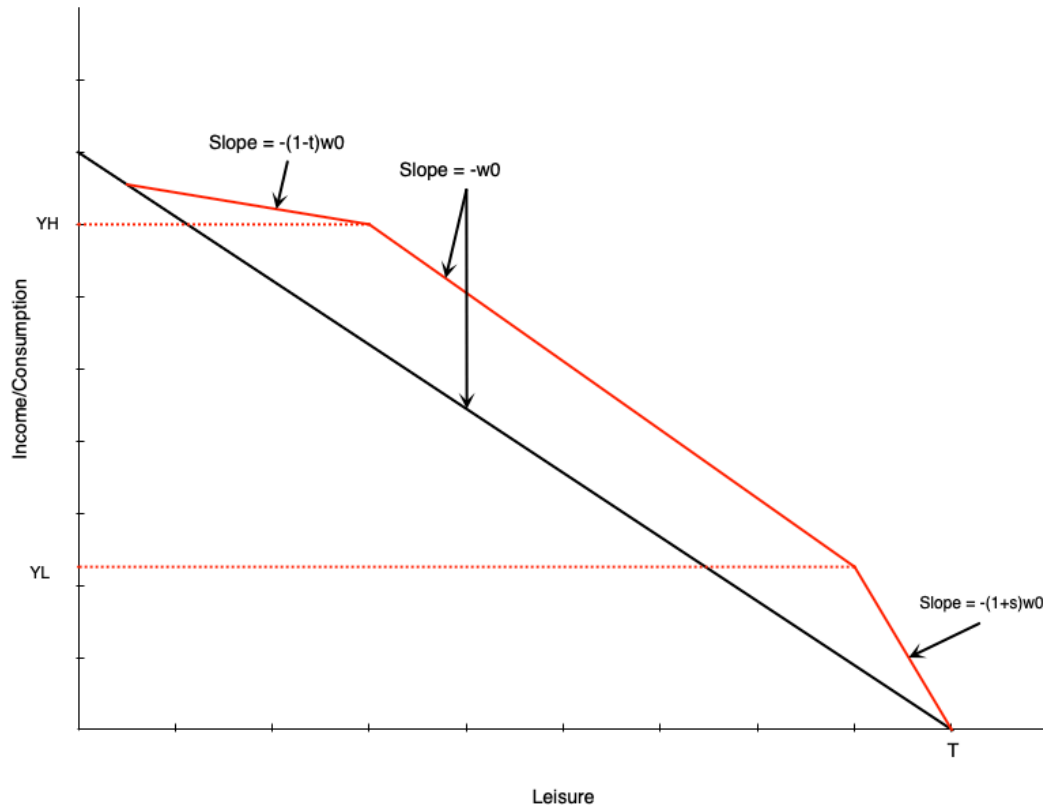
Two types of tax credits:

1. Non-refundable: Can reduce tax owed to zero (but not below)
2. Refundable: Can reduce tax owed below zero (results in payment)

## Income Maintenance Programs via Refundable Tax Credits

- [Canada Worker Benefit \(CWB\)](#) - Canada
- [Earned Income Tax Credit \(EITC\)](#) - US

# Tax Credits: Budget Constraint



Three segments:

1. **Phase-in**: Subsidy proportional to earned income
2. **Plateau**: Maximum benefit level
3. **Phase-out**: Benefit reduced with earned income

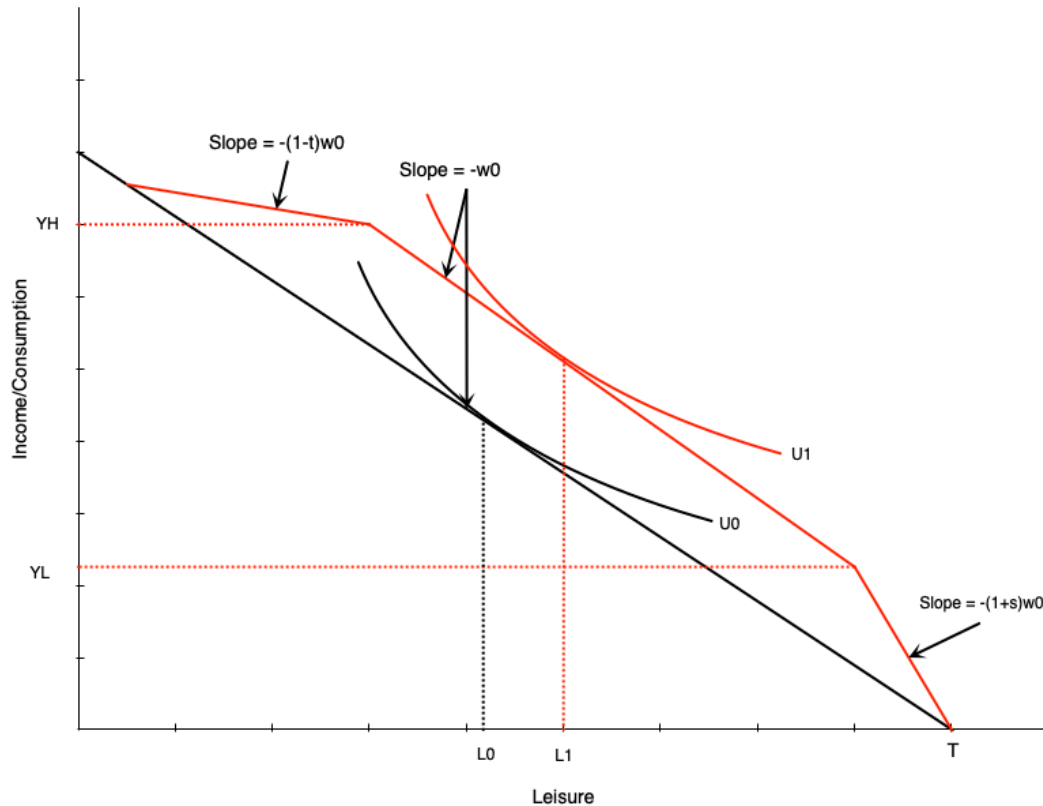
Combines elements of:

- Wage subsidy (phase-in)
- Demogrant (plateau)
- NIT (phase-out)

# Tax Credits: Effect in Plateau Range

Person initially in middle range:

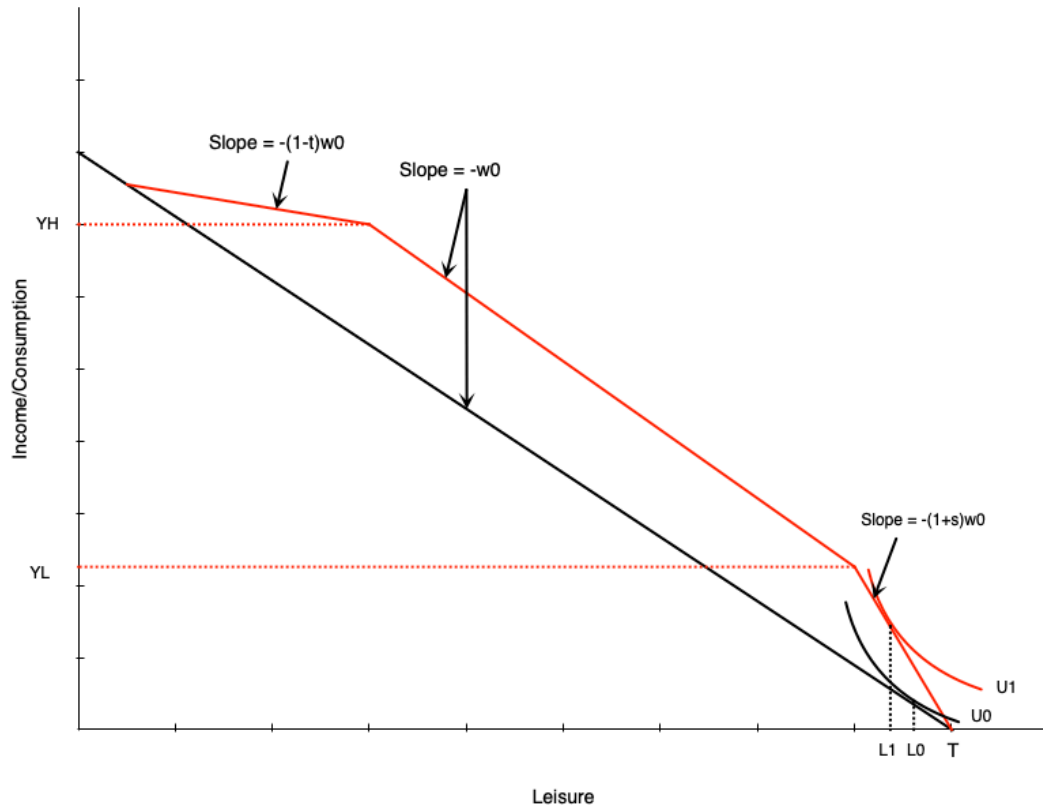
- Tax credit gives maximum benefit
- Acts like **pure income effect**
- If leisure is normal good  $\rightarrow$  reduce work hours



# Tax Credits: Effect in Phase-In Range

Person in phase-in range:

- Sees increase in effective wage
- Acts like **pure wage subsidy**
- Income and substitution effects in opposite directions



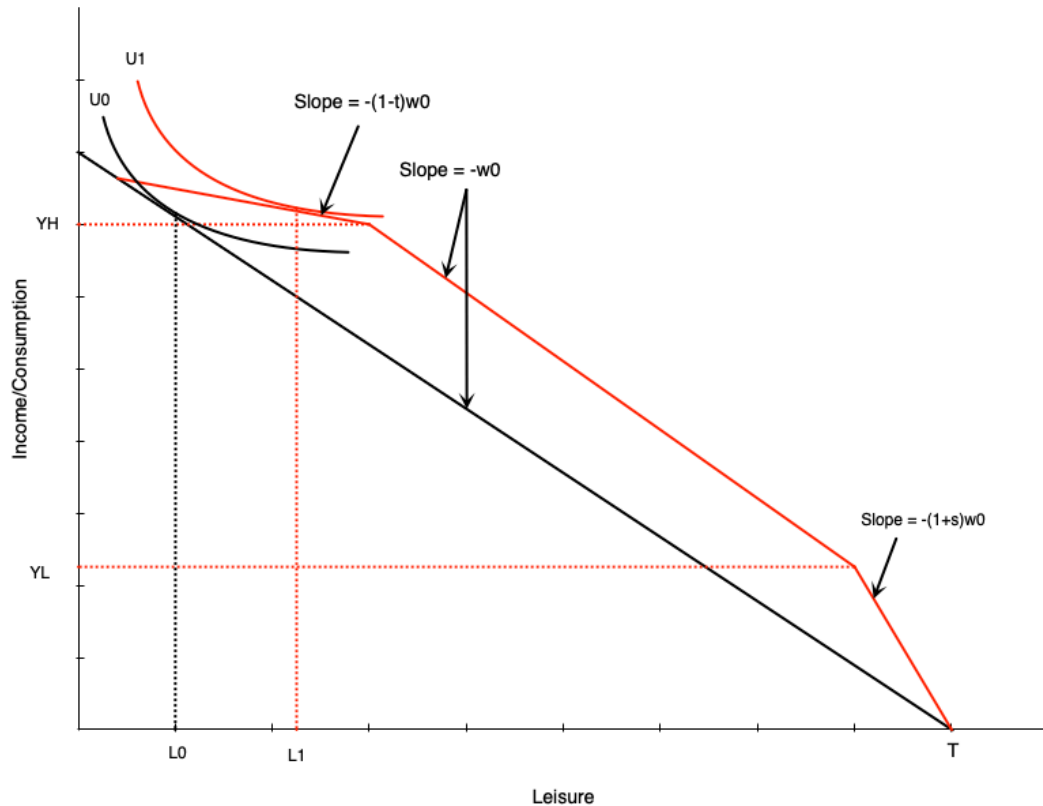
## Warning

Effect on hours is **ambiguous**

# Tax Credits: Effect in Phase-Out Range

Person in phase-out range:

- Similar to **negative income tax**
- Income effect  $\rightarrow$  reduce hours  $\downarrow$
- Substitution effect  $\rightarrow$  reduce hours  $\downarrow$



❗ Important

Effect on work is **unambiguously negative**



# Unemployment Insurance

# Employment Insurance: Overview

## Purpose:

- Safety net for workers who lose their job
- Collect income while unemployed

## Funding:

- Payroll taxes on employers and employees

## Two main components:

1. Regular benefits - for those unemployed through no fault of their own
2. Special benefits - sickness, maternity, parental leave, caregiving

# Employment Insurance: Benefits

## Regular Benefits Structure

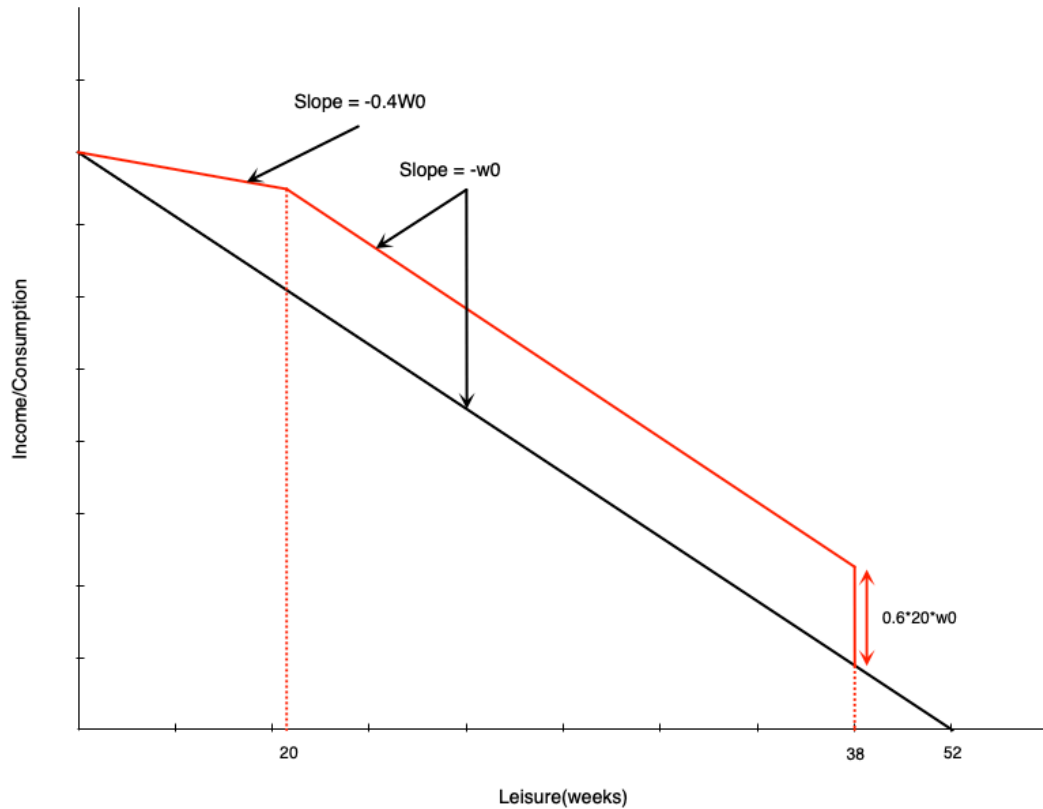
- 55% of average insurable weekly earnings
- Up to a maximum amount
- Duration: 14 to 45 weeks
  - Depends on unemployment rate and hours worked in last year

# EI: Simplified Model Parameters

To analyze with income-leisure diagram:

- Must work **14 weeks** in the year to qualify
- Benefit is **60% of weekly wage**
- Maximum of **20 weeks** of benefits
- Can work up to 32 weeks and get full benefits
- Each week worked beyond 32 → lose benefits for that week
- 60% benefit = **60% tax** on earnings beyond 32 weeks
- **Cannot work and collect EI simultaneously**

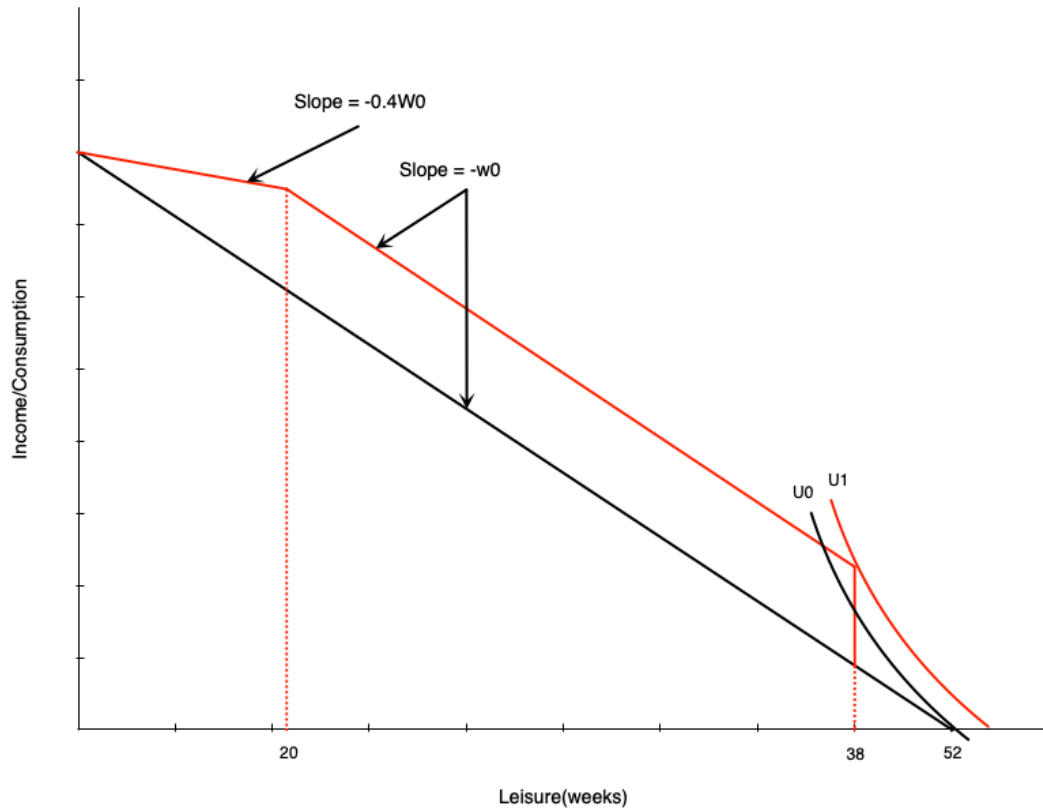
# El: Budget Constraint



## Three segments:

1. First 14 weeks: Paid at usual wage  $w_0$
2. Weeks 14-32 of work: Get 20 weeks of benefits
  - Benefit =  $0.6 \times 20 \times w_0$
3. More than 32 weeks of work: Benefits reduced week-for-week
  - Benefits paid at 60% of wage
  - Work paid at 100% of wage
  - **Keep 40% of wage** in this range

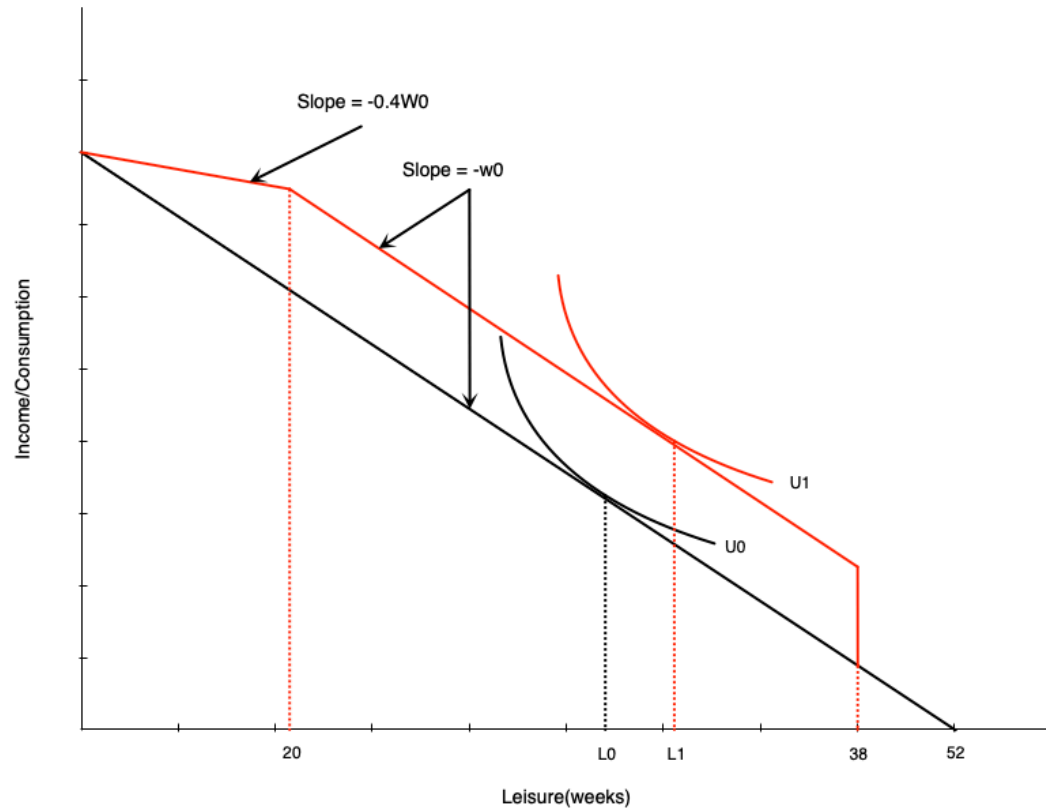
# EI: Effect on Non-Participants



## Effect on Non-Workers

A non-participant might start working due to large increase in non-labour income after 14 hours of work. Induces labour force entry.

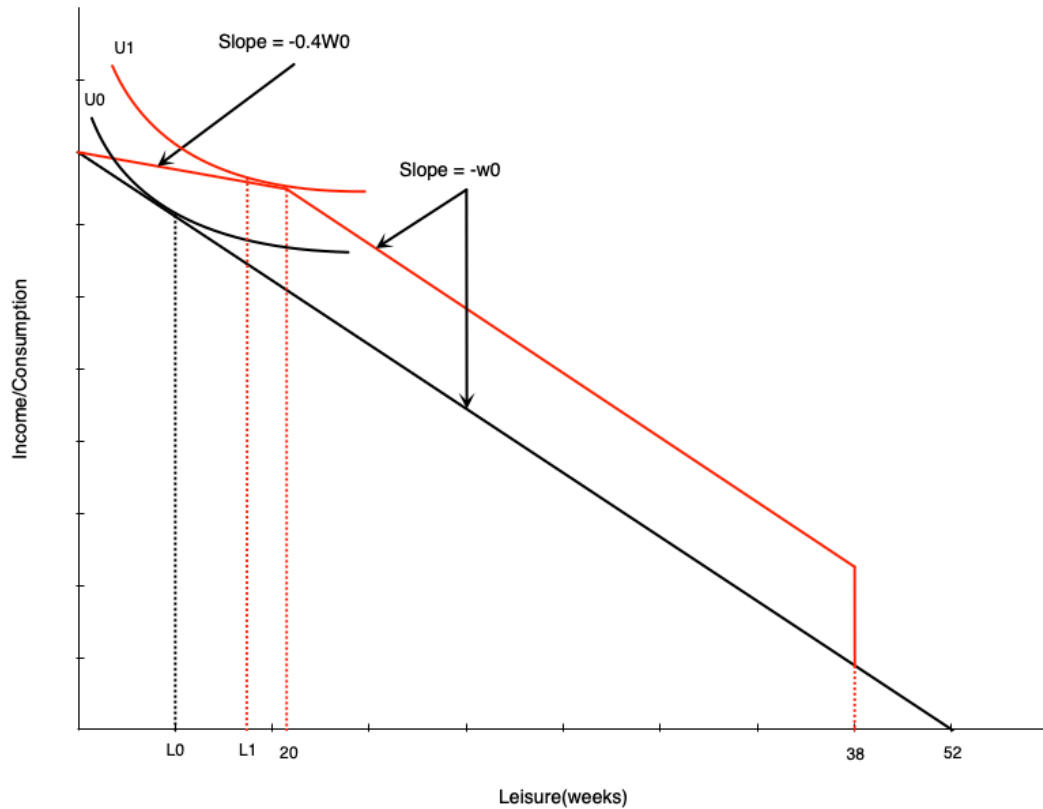
# EI: Effect in Middle Range



Person getting full benefits:

- Acts like pure income effect
- Will reduce work weeks

# EI: Effect on High-Hour Workers



In phase-out range:

- Acts like **negative income tax**
- Income effect → reduce hours ↓
- Substitution effect → reduce hours ↓

⚠ Important

Effect **unambiguously negative**



# Evidence: Maine vs. New Brunswick

## Natural experiment:

- Initially had very similar EI schemes
- Later, New Brunswick increased benefit generosity relative to Maine
- **Difference-in-differences approach:**
  - Compare changes in NB before/after to Maine over same period
  - Maine acts as control group
  - Controls for other time-varying factors

# Evidence: Maine vs. New Brunswick Results

## New Brunswick experienced:

- Higher labour force participation for marginally-attached workers
- Lower hours for those with stronger attachments

### Interpretation

- Corresponds exactly to model predictions
- Highlights work disincentive effects of income maintenance programs

# Disability and Workers Compensation

# Introduction

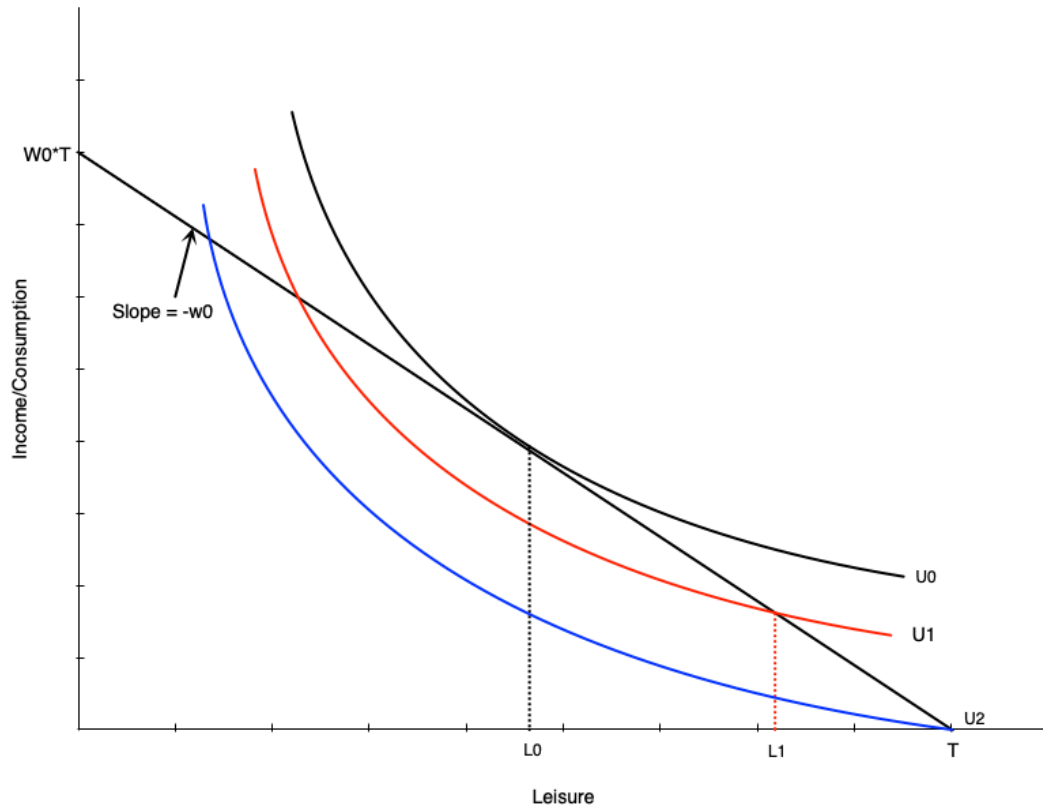
## What are these programs?

- Provide income support to injured/ill workers
- Come from public and private sources
- Premiums paid by employers and employees

### Work Incentive Considerations

- Severe injury/disability: work incentives not a concern (unable to work)
- Partial disability: people may still be able to work

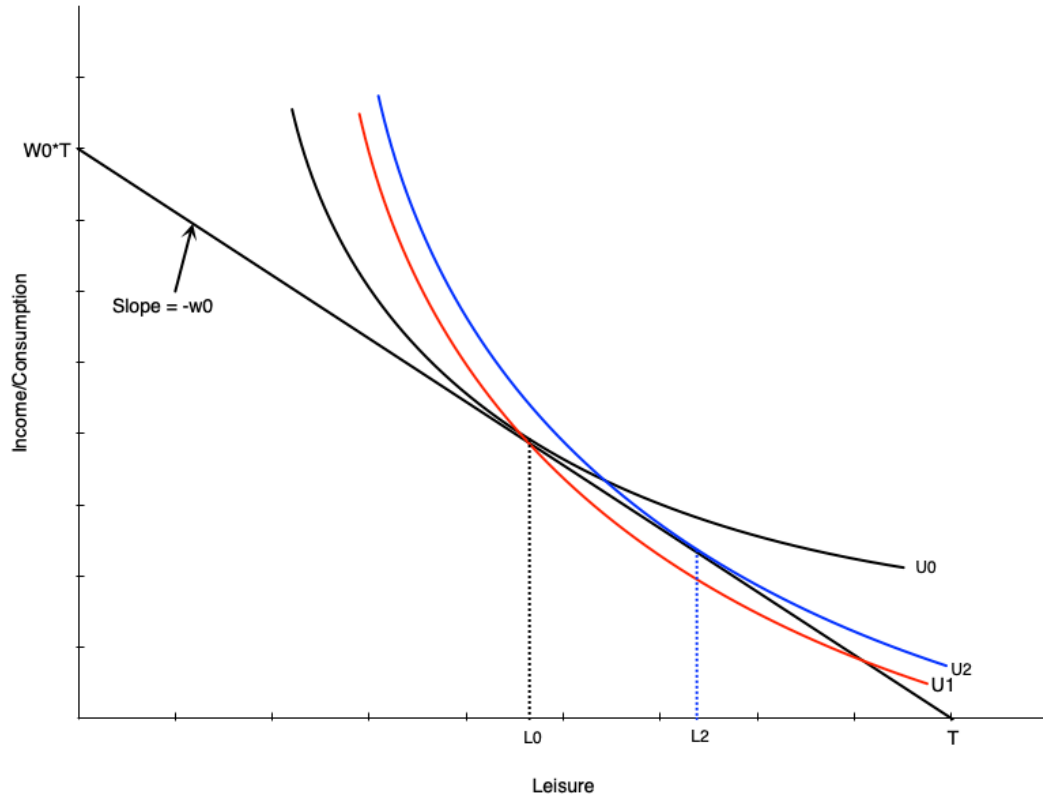
# Effect of Disability on Labour Supply



Three scenarios:

1. Initially: Choose  $L_0$  leisure, work  $H_0$  hours
2. **Injured but can work:** Choose  $L_1$  leisure, work  $H_1$  hours
  - Lower (non-optimal) indifference curve
3. **Completely disabled:** Choose  $T$  leisure, zero work
  - Even lower indifference curve

# Disability and Preferences



Disability may change preferences:

- Work becomes more difficult
- Preferences tilt toward leisure
- Indifference curve becomes steeper
- MRS increases at each point
- More willing to give up consumption for leisure

Result:

- Even if unconstrained, choose more leisure
- Person moves from  $L_0$  to  $L_2$

# Compensation Schemes

## Typical disability insurance structure:

- Pays a fraction of previous earnings

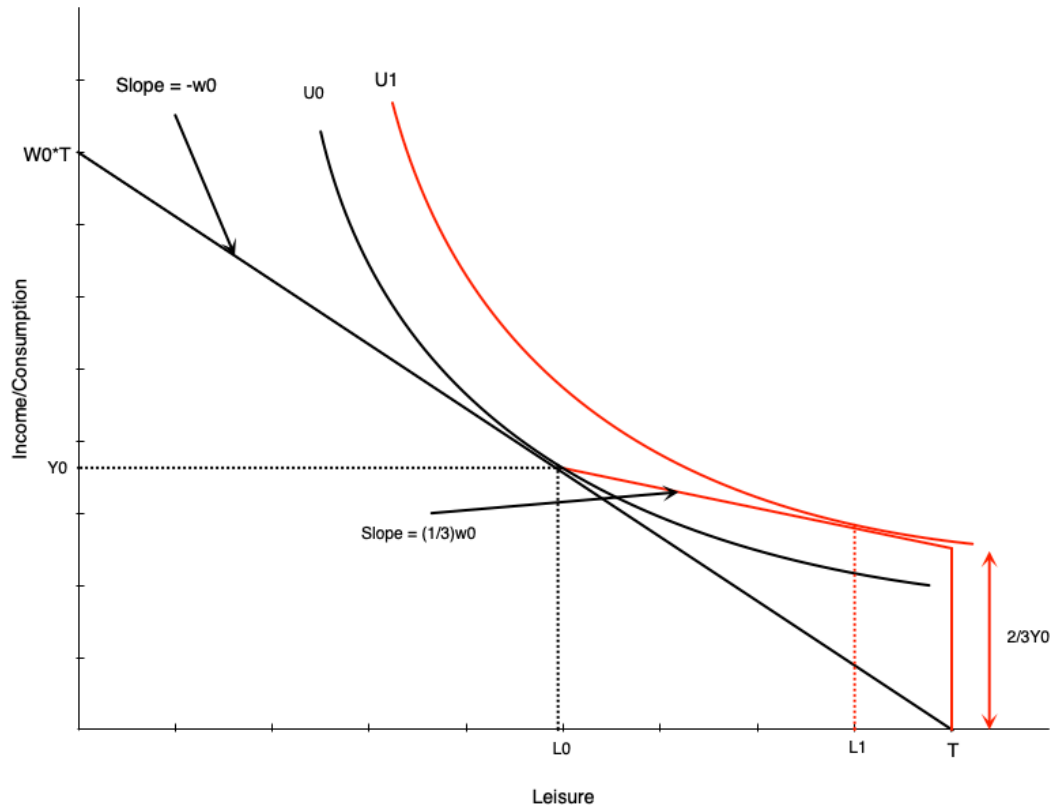
### Example: Ontario WSIB

- 85% of net earnings (up to maximum)
- Lasts until return to work or age 65

## Two schemes we'll examine:

1. Two-thirds income replacement
2. Full income replacement

# Two-Thirds Compensation



Looks like negative income tax:

- At zero hours: get 2/3 of previous income
- For each hour worked:
  - Gain full wage
  - Lose 2/3 of wage in benefits
  - **Keep 1/3 of wage**

For workers with flexible hours:

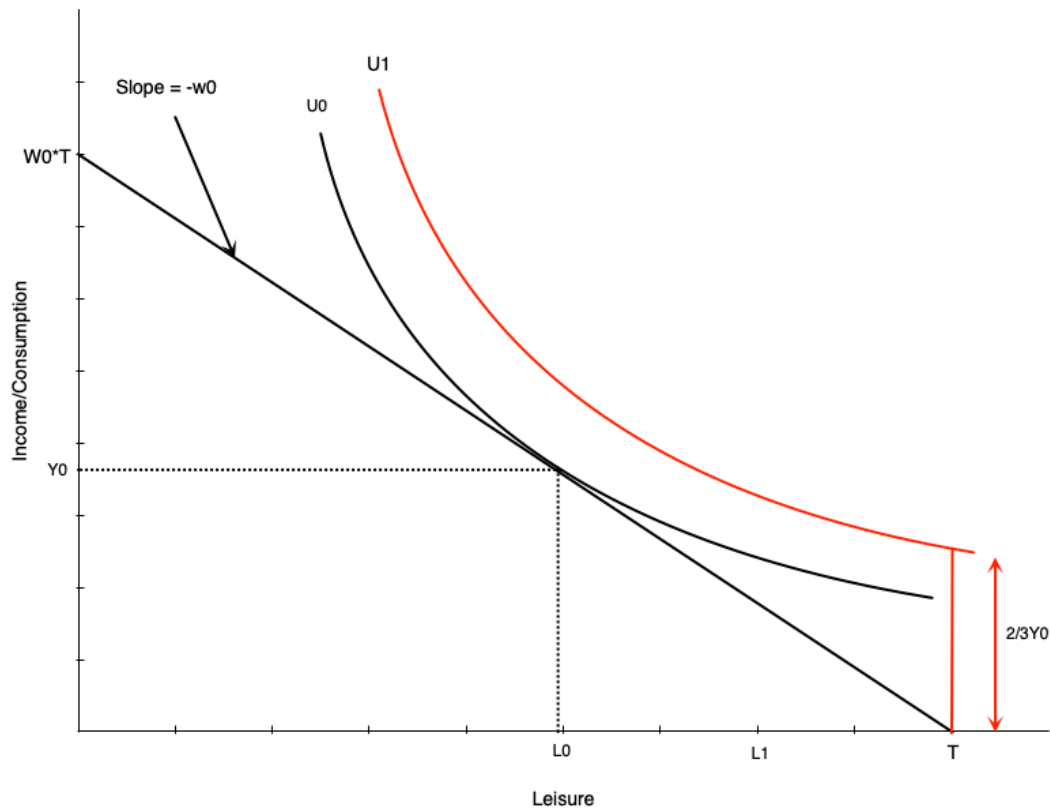
- Effects match NIT
- Negative effect on hours worked



# Restricted Hours: Generous Compensation

What if worker can only work  $H_0$  hours or zero?

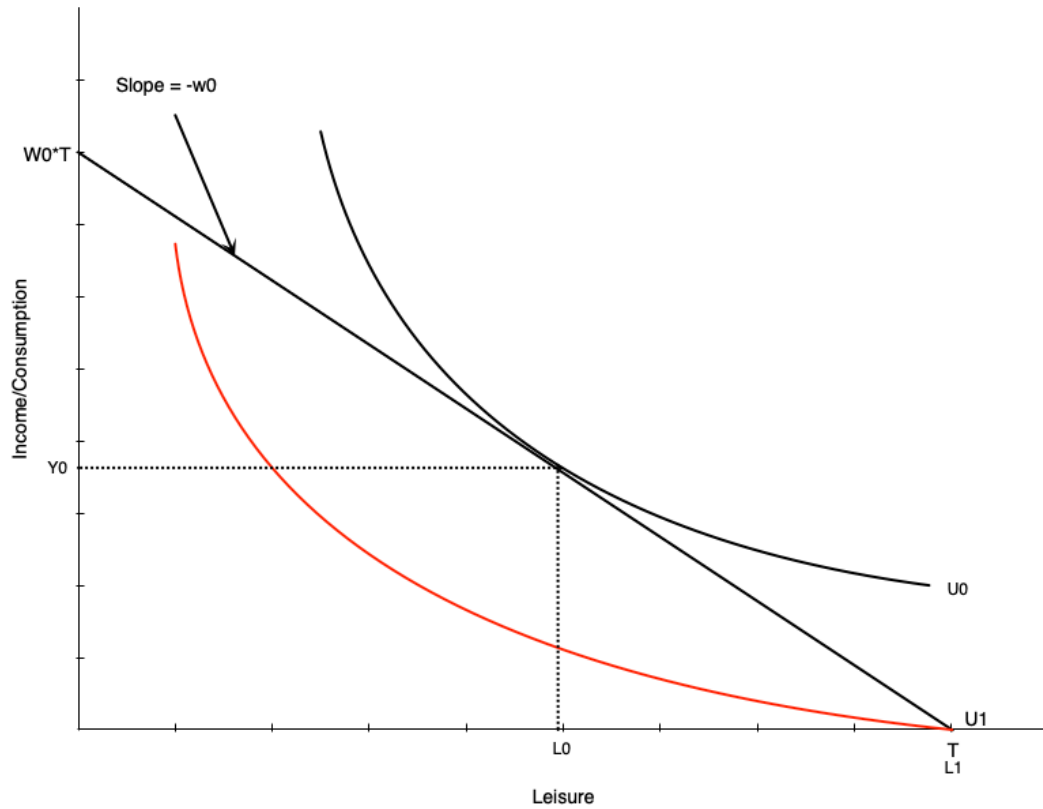
- If scheme generous enough  $\rightarrow$  choose zero hours
- Utility at zero hours  $>$  utility at  $H_0$  hours



## ⚠ Implication

Strong work disincentives of generous scheme

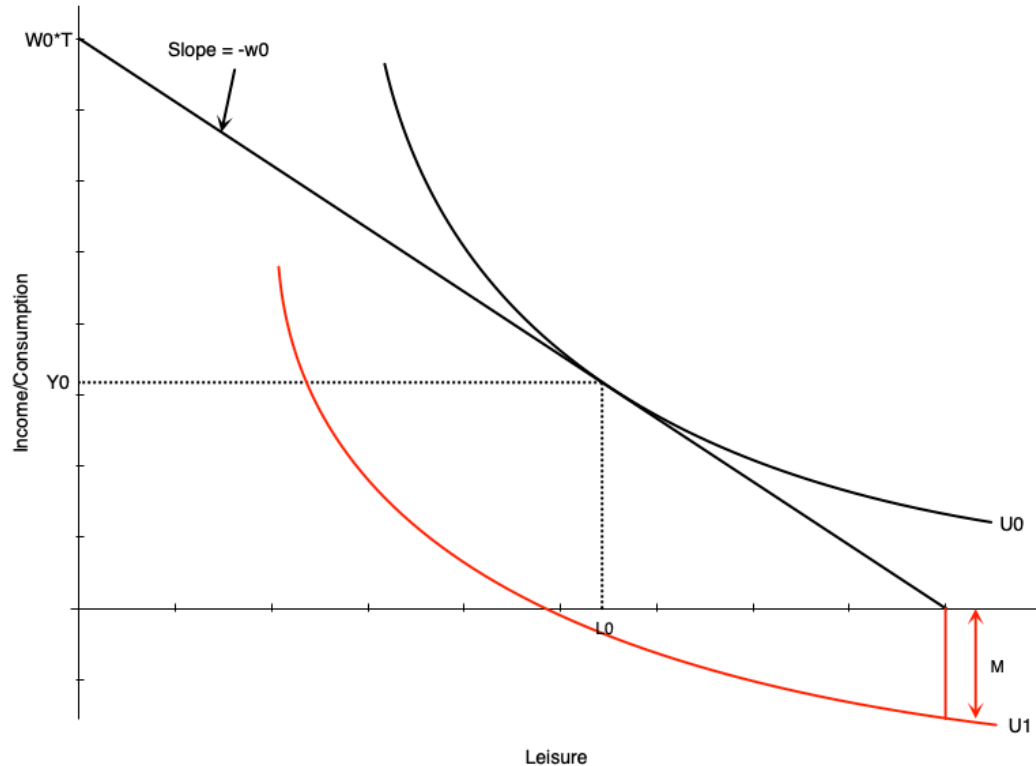
# No Compensation Case



## ! Not Providing Compensation is Problematic

- Person who can work chooses  $H_0$  hours
- Person who cannot work gets lower utility
- End up substantially worse off

# Medical Expenses Case

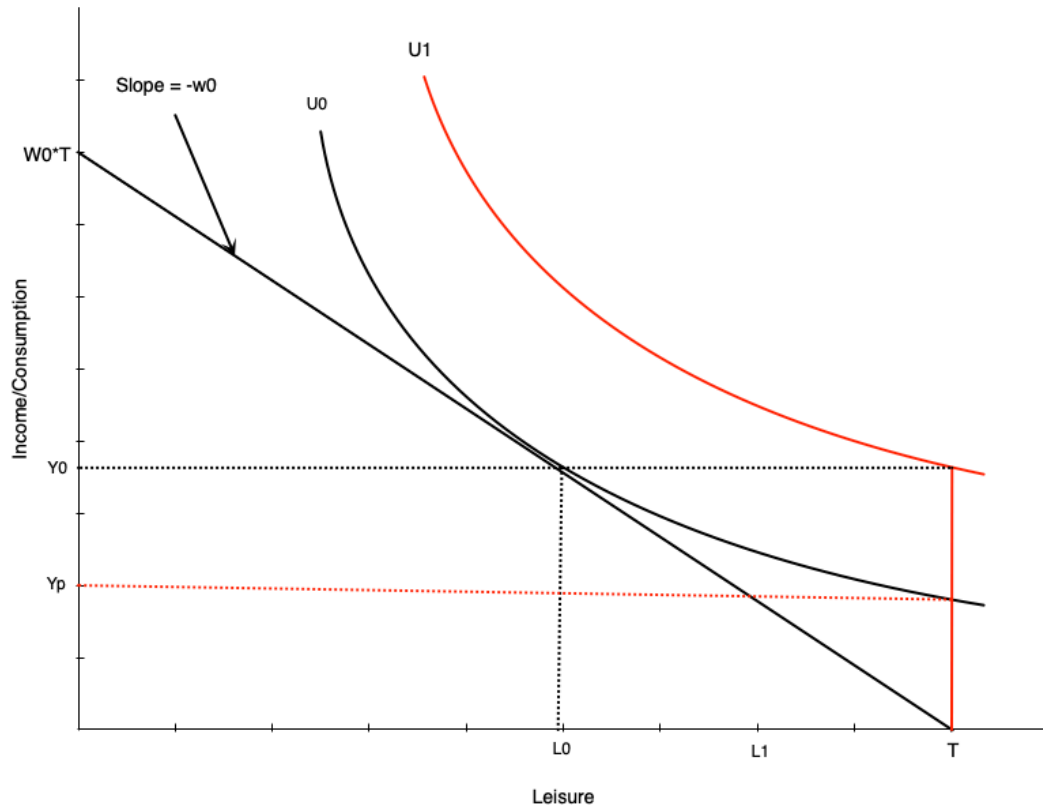


Medical expenses worsen situation:

- Act like reduction in non-labour income
- If large enough, non-labour income could go negative
- Occurs with no/very low compensation

This is an extreme scenario

# Optimal Compensation Level



How much to compensate non-workers?

Two options:

1. Full income replacement

- Higher indifference curve ( $U_1$ ) than working

2. Same satisfaction as working

- Same indifference curve as working
- Lower compensation ( $Y_p$ )

*i* Note

The optimal level is a policy question

# Child Care Subsidies

# Introduction

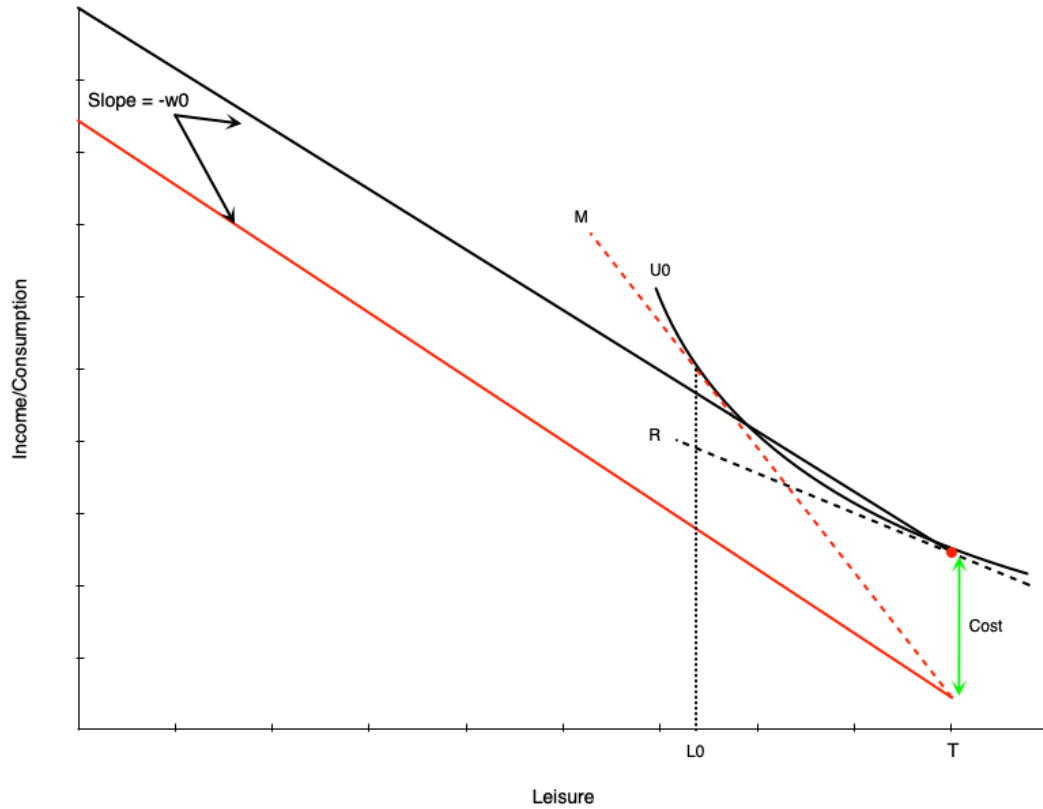
## Canada's child care expansion:

- Goal: reduce average cost to **\$10/day by 2026**
- More than half of provinces already at \$10/day average
- Subsidy varies substantially by province
- Requires provincial-federal agreements

### Quebec Case Study

Had \$10/day child care since 1997. Provides useful evidence on effects.

# Child Care Costs: Non-Working Case



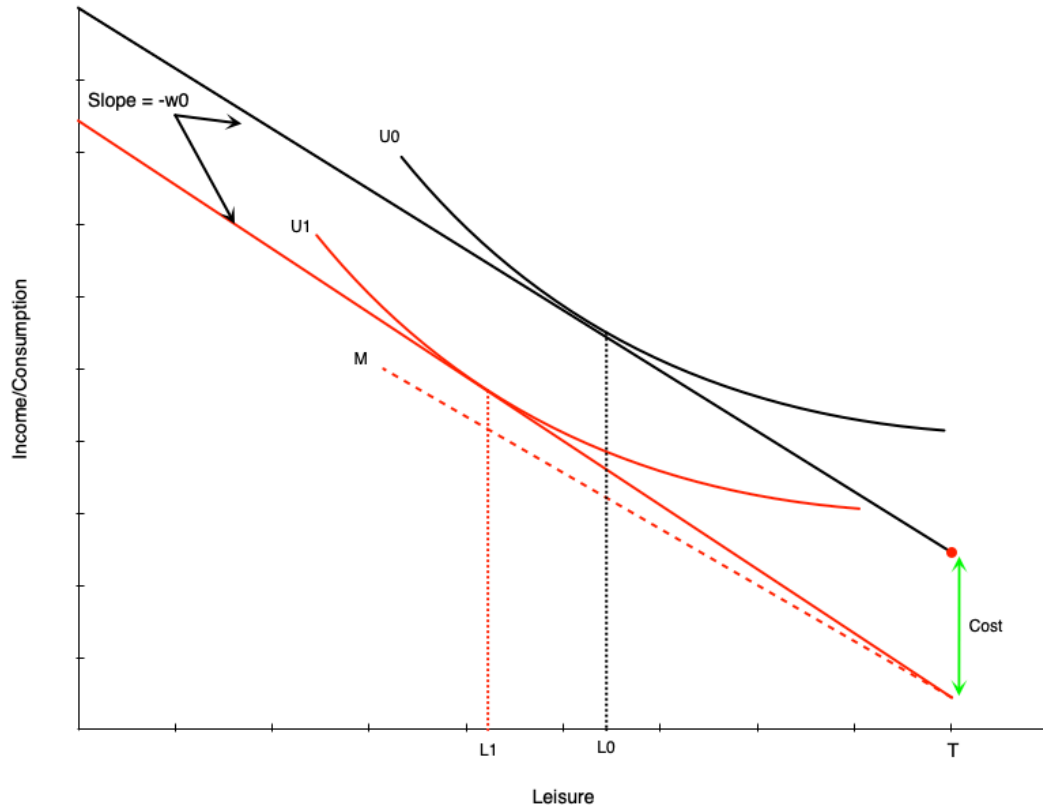
Budget line becomes discontinuous:

- At zero hours: red dot
- For any work: drops by child care costs
- Then straight line with slope = wage

Effect:

- Child care costs **increase reservation wage**
- Need higher wage to induce work
- This person chooses not to work

# Child Care Costs: Working Case

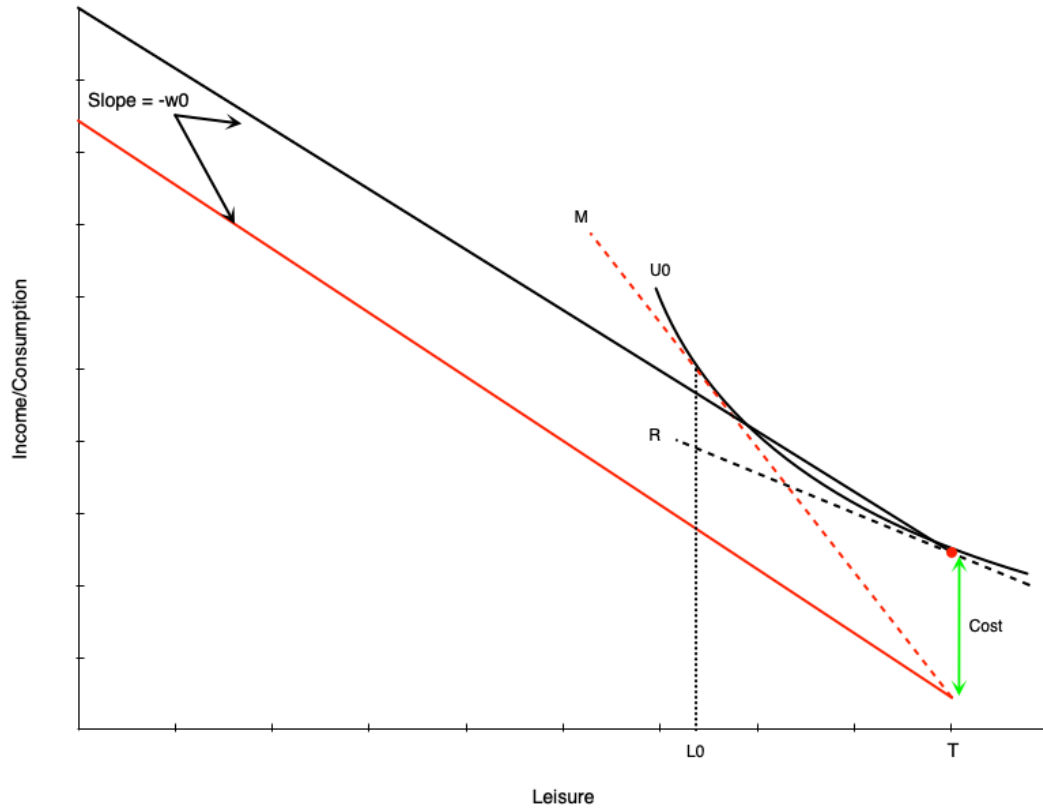


Person with reservation wage above wage + costs:

- Works positive hours
- Child care costs = **negative pure income effect**
- Choose less leisure, more work
- Must work more to offset income loss



# Child Care Costs: Labour Supply Gap



## ! Creates Gap in Labour Supply Schedule

- Person who works will work minimum of  $H_0 = T - L_0$  hours
- Not worth working less due to large child care costs
- Labour supply schedule jumps from zero to  $H_0$  hours

# Child Care Subsidies: Effects

A full subsidy (costs  $\rightarrow$  zero) would:

1. Move budget line up in parallel
2. Reduce reservation wage
  - Lower barrier  $\rightarrow$  encourages participation
  - Especially encourages part-time work
3. Create positive income effect
  - Increases consumption of leisure
  - Reduces work hours

# Evidence: Quebec's \$5/Day Program

## Program details:

- Implemented in 2000
- Currently sliding scale: \$8.25 to \$21.45

## Research design:

- Difference-in-differences approach
- Compared Quebec mothers to mothers in other provinces
- Other provinces = control group

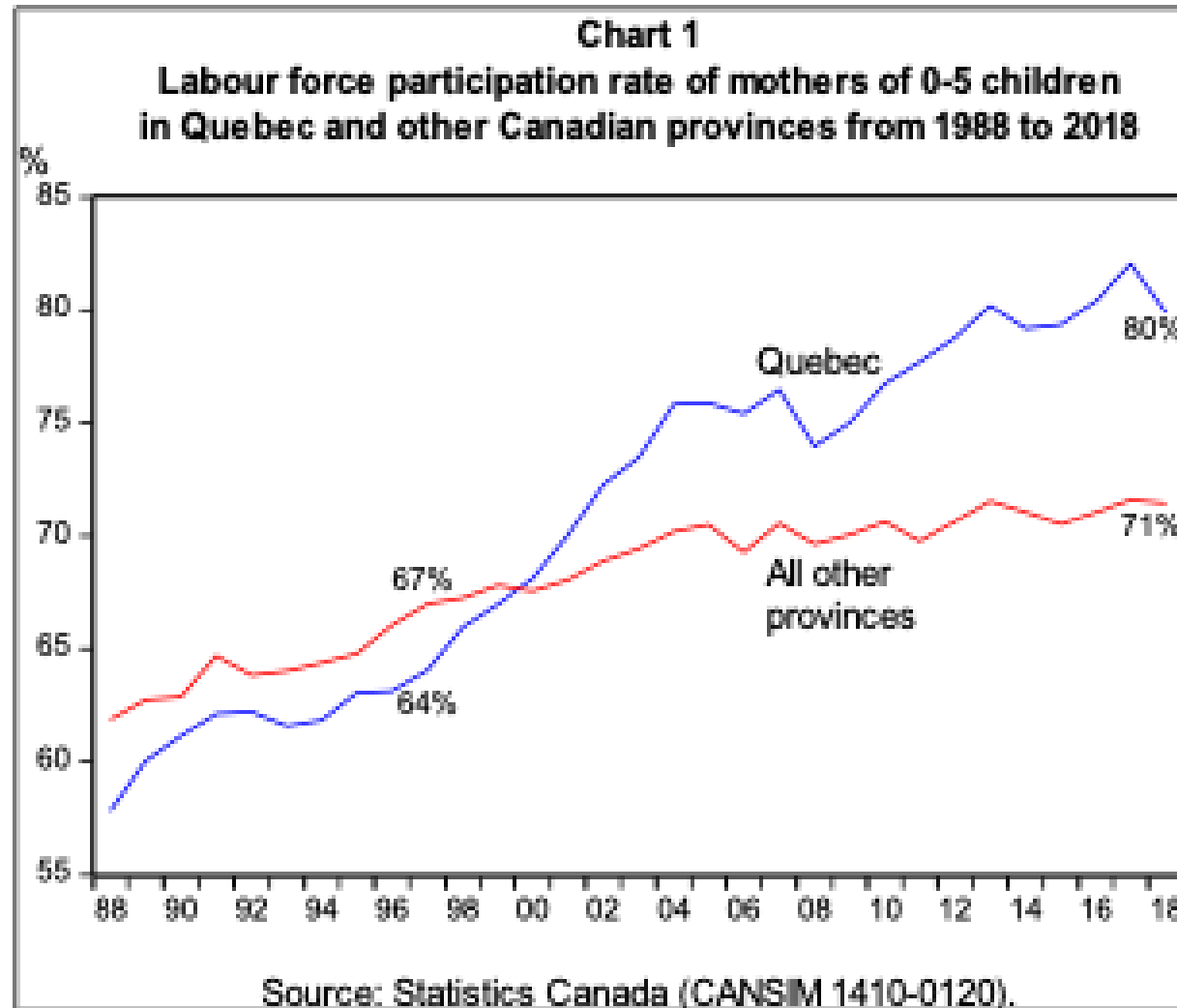
### Key Finding

Labour force participation of mothers with young children ↑ **7.7 percentage points**

### Warning

Unfortunately also found negative effects on child mental health, non-cognitive behaviour, crime later in life

# Evidence: Visual Summary



Source: Fortin (2019)

# Summary

# Key Takeaways

## Income maintenance programs:

- Provide crucial income support to various groups
- Can create work disincentives in some cases
- Must balance support needs against incentive effects

## We examined several programs through the labour supply model:

- Demogrants, welfare, NIT, wage subsidies, tax credits
- Employment insurance
- Disability and workers compensation
- Child care subsidies